In this chapter we will develop the theory of differentiation. Integration theory, including the fundamental theorem that relates differentiation and integration, will be the subject of the next chapter. We will assume that the reader is already familiar with the geometrical and physical interpretations of the derivative of a function as described in introductory calculus courses. Consequently, we will concentrate on the mathematical aspects of the derivative and not go into its applications in geometry, physics, economics, and so on.

The first section is devoted to a presentation of the basic results concerning the differentiation of functions. In Section 6.2 we discuss the fundamental Mean Value Theorem and some of its applications. In Section 6.3 the important L'Hospital Rules are presented for the calculation of certain types of "indeterminate" limits.

In Section 6.4 we give a brief discussion of Taylor's Theorem and a few of its applications—for example, to convex functions and to Newton's Method for the location of roots.

Section 6.1 The Derivative

In this section we will present some of the elementary properties of the derivative. We begin with the definition of the derivative of a function.

6.1.1 Definition Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, and let $c \in I$. We say that a real number L is the **derivative of f at c** if given any $\varepsilon > 0$ there exists $\delta(\varepsilon) > 0$ such that if $x \in I$ satisfies $0 < |x - c| < \delta(\varepsilon)$, then

$$\left|\frac{f(x) - f(c)}{x - c} - L\right| < \varepsilon. \tag{1}$$

In this case we say that f is **differentiable** at c, and we write $f'(c)$ for L.

In other words, the derivative of f at c is given by the limit

$$f'(c) = \lim_{x \to c} \frac{f(x) - f(c)}{x - c} \tag{2}$$

provided this limit exists. (We allow the possibility that c may be the endpoint of the interval.)

Note It is possible to define the derivative of a function having a domain more general than an interval (since the point c need only be an element of the domain and also a cluster point of the domain) but the significance of the concept is most naturally apparent for functions defined on intervals. Consequently we shall limit our attention to such functions.

Whenever the derivative of $f : I \to \mathbb{R}$ exists at a point $c \in I$, its value is denoted by $f'(c)$. In this way we obtain a function f' whose domain is a subset of the domain of f. In working with the function f', it is convenient to regard it also as a function of x. For example, if $f(x) := x^2$ for $x \in \mathbb{R}$, then at any c in $\mathbb{R}$ we have

$$f'(c) = \lim_{x \to c} \frac{f(x) - f(c)}{x - c} = \lim_{x \to c} \frac{x^2 - c^2}{x - c} = \lim_{x \to c} (x + c) = 2c.$$

Thus, in this case, the function f' is defined on all of $\mathbb{R}$ and $f'(x) = 2x$ for $x \in \mathbb{R}$.

We now show that continuity of f at a point c is a necessary (but not sufficient) condition for the existence of the derivative at c.

6.1.2 Theorem *If $f : I \to \mathbb{R}$ has a derivative at $c \in I$, then f is continuous at c.*

Proof. For all $x \in I$, $x \neq c$, we have

$$f(x) - f(c) = \left(\frac{f(x) - f(c)}{x - c}\right)(x - c).$$

Since $f'(c)$ exists, we may apply Theorem 4.2.4 concerning the limit of a product to conclude that

$$\begin{aligned}\lim_{x \to c}(f(x) - f(c)) &= \lim_{x \to c}\left(\frac{f(x) - f(c)}{x - c}\right)\left(\lim_{x \to c}(x - c)\right) \\ &= f'(c) \cdot 0 = 0.\end{aligned}$$

Therefore, $\lim_{x \to c} f(x) = f(c)$ so that f is continuous at c. Q.E.D.

The continuity of $f : I \to \mathbb{R}$ at a point does not assure the existence of the derivative at that point. For example, if $f(x) := |x|$ for $x \in \mathbb{R}$, then for $x \neq 0$ we have $(f(x) - f(0))/(x - 0) = |x|/x$, which is equal to 1 if $x > 0$, and equal to -1 if $x < 0$. Thus the limit at 0 does not exist [see Example 4.1.10(b)], and therefore the function is not differentiable at 0. Hence, continuity at a point c is *not* a sufficient condition for the derivative to exist at c.

Remark By taking simple algebraic combinations of functions of the form $x \mapsto |x - c|$, it is not difficult to construct continuous functions that do not have a derivative at a finite (or even a countable) number of points. In 1872, Karl Weierstrass astounded the mathematical world by giving an example of a function that is *continuous at every point but whose derivative does not exist anywhere.* Such a function defied geometric intuition about curves and tangent lines, and consequently spurred much deeper investigations into the concepts of real analysis. It can be shown that the function f defined by the series

$$f(x) := \sum_{n=0}^{\infty} \frac{1}{2^n} \cos(3^n x)$$

has the stated property. A very interesting historical discussion of this and other examples of continuous, nondifferentiable functions is given in Kline, pp. 955–966, and also in Hawkins, pp. 44–46. A detailed proof for a slightly different example can be found in Appendix E.

There are a number of basic properties of the derivative that are very useful in the calculation of the derivatives of various combinations of functions. We now provide the justification of some of these properties, which will be familiar to the reader from earlier courses.

6.1.3 Theorem *Let $I \subseteq \mathbb{R}$ be an interval, let $c \in I$, and let $f : I \to \mathbb{R}$ and $g : I \to \mathbb{R}$ be functions that are differentiable at c. Then:*

(a) *If $\alpha \in \mathbb{R}$, then the function αf is differentiable at c, and*

$$(\alpha f)'(c) = \alpha f'(c). \tag{3}$$

(b) *The function $f+g$ is differentiable at c, and*

$$(f+g)'(c) = f'(c) + g'(c). \tag{4}$$

(c) (*Product Rule*) *The function fg is differentiable at c, and*

$$(fg)'(c) = f'(c)g(c) + f(c)g'(c). \tag{5}$$

(d) (*Quotient Rule*) *If $g(c) \neq 0$, then the function f/g is differentiable at c, and*

$$\left(\frac{f}{g}\right)'(c) = \frac{f'(c)g(c) - f(c)g'(c)}{(g(c))^2}. \tag{6}$$

Proof. We shall prove (c) and (d), leaving (a) and (b) as exercises for the reader.

(c) Let $p := fg$; then for $x \in I$, $x \neq c$, we have

$$\begin{aligned}\frac{p(x)-p(c)}{x-c} &= \frac{f(x)g(x) - f(c)g(c)}{x-c} \\ &= \frac{f(x)g(x) - f(c)g(x) + f(c)g(x) - f(c)g(c)}{x-c} \\ &= \frac{f(x)-f(c)}{x-c}\cdot g(x) + f(c)\cdot\frac{g(x)-g(c)}{x-c}.\end{aligned}$$

Since g is continuous at c, by Theorem 6.1.2, then $\lim_{x\to c} g(x) = g(c)$. Since f and g are differentiable at c, we deduce from Theorem 4.2.4 on properties of limits that

$$\lim_{x\to c}\frac{p(x)-p(c)}{x-c} = f'(c)g(c) + f(c)g'(c).$$

Hence $p := fg$ is differentiable at c and (5) holds.

(d) Let $q := f/g$. Since g is differentiable at c, it is continuous at that point (by Theorem 6.1.2). Therefore, since $g(c) \neq 0$, we know from Theorem 4.2.9 that there exists an interval $J \subseteq I$ with $c \in J$ such that $g(x) \neq 0$ for all $x \in J$. For $x \in J$, $x \neq c$, we have

$$\begin{aligned}\frac{q(x)-q(c)}{x-c} &= \frac{f(x)/g(x) - f(c)/g(c)}{x-c} = \frac{f(x)g(c) - f(c)g(x)}{g(x)g(c)(x-c)} \\ &= \frac{f(x)g(c) - f(c)g(c) + f(c)g(c) - f(c)g(x)}{g(x)g(c)(x-c)} \\ &= \frac{1}{g(x)g(c)}\left[\frac{f(x)-f(c)}{x-c}\cdot g(c) - f(c)\cdot\frac{g(x)-g(c)}{x-c}\right].\end{aligned}$$

Using the continuity of g at c and the differentiability of f and g at c, we get

$$q'(c) = \lim_{x\to c}\frac{q(x)-q(c)}{x-c} = \frac{f'(c)g(c) - f(c)g'(c)}{(g(c))^2}.$$

Thus, $q = f/g$ is differentiable at c and equation (6) holds. Q.E.D.

Mathematical Induction may be used to obtain the following extensions of the differentiation rules.

6.1.4 Corollary *If $f_1, f_2, \ldots, f_n$ are functions on an interval I to $\mathbb{R}$ that are differentiable at $c \in I$, then:*

(a) *The function $f_1 + f_2 + \cdots + f_n$ is differentiable at c and*

$$(f_1 + f_2 + \cdots + f_n)'(c) = f_1'(c) + f_2'(c) + \cdots + f_n'(c). \tag{7}$$

(b) *The function $f_1 f_2 \cdots f_n$ is differentiable at c, and*

$$(f_1 f_2 \cdots f_n)'(c) = f_1'(c) f_2(c) \cdots f_n(c) + f_1(c) f_2'(c) \cdots f_n(c) + \cdots + f_1(c) f_2(c) \cdots f_n'(c). \tag{8}$$

An important special case of the extended product rule (8) occurs if the functions are equal, that is, $f_1 = f_2 = \cdots = f_n = f$. Then (8) becomes

$$(f^n)'(c) = n(f(c))^{n-1} f'(c). \tag{9}$$

In particular, if we take $f(x) := x$, then we find the derivative of $g(x) := x^n$ to be $g'(x) = nx^{n-1}$, $n \in \mathbb{N}$. The formula is extended to include negative integers by applying the Quotient Rule 6.1.3(d).

Notation If $I \subseteq \mathbb{R}$ is an interval and $f : I \to \mathbb{R}$, we have introduced the notation f' to denote the function whose domain is a subset of I and whose value at a point c is the derivative $f'(c)$ of f at c. There are other notations that are sometimes used for f'; for example, one sometimes writes Df for f'. Thus one can write formulas (4) and (5) in the form:

$$D(f+g) = Df + Dg, \qquad D(fg) = (Df) \cdot g + f \cdot (Dg).$$

When x is the "independent variable," it is common practice in elementary courses to write df/dx for f'. Thus formula (5) is sometimes written in the form

$$\frac{d}{dx}(f(x)g(x)) = \left(\frac{df}{dx}(x)\right) g(x) + f(x) \left(\frac{dg}{dx}(x)\right).$$

This last notation, due to Leibniz, has certain advantages. However, it also has certain disadvantages and must be used with some care.

The Chain Rule

We now turn to the theorem on the differentiation of composite functions known as the "Chain Rule." It provides a formula for finding the derivative of a composite function $g \circ f$ in terms of the derivatives of g and f.

We first establish the following theorem concerning the derivative of a function at a point that gives us a very nice method for proving the Chain Rule. It will also be used to derive the formula for differentiating inverse functions.

6.1.5 Carathéodory's Theorem *Let f be defined on an interval I containing the point c. Then f is differentiable at c if and only if there exists a function φ on I that is continuous at c and satisfies*

$$f(x) - f(c) = \varphi(x)(x - c) \quad \text{for} \quad x \in I. \tag{10}$$

In this case, we have $\varphi(c) = f'(c)$.

Proof. ($\Rightarrow$) If $f'(c)$ exists, we can define φ by

$$\varphi(x) := \begin{cases} \dfrac{f(x) - f(c)}{x - c} & \text{for} \quad x \neq c,\ x \in I, \\ f'(c) & \text{for} \quad x = c. \end{cases}$$

The continuity of φ follows from the fact that $\lim_{x \to c} \varphi(x) = f'(c)$. If $x = c$, then both sides of (10) equal 0, while if $x \neq c$, then multiplication of $\varphi(x)$ by $x - c$ gives (10) for all other $x \in I$.

($\Leftarrow$) Now assume that a function φ that is continuous at c and satisfying (10) exists. If we divide (10) by $x - c \neq 0$, then the continuity of φ implies that

$$\varphi(c) = \lim_{x \to c} \varphi(x) = \lim_{x \to c} \frac{f(x) - f(c)}{x - c}$$

exists. Therefore f is differentiable at c and $f'(c) = \varphi(c)$. Q.E.D.

To illustrate Carathéodory's Theorem, we consider the function f defined by $f(x) := x^3$ for $x \in \mathbb{R}$. For $c \in \mathbb{R}$, we see from the factorization

$$x^3 - c^3 = (x^2 + cx + c^2)(x - c)$$

that $\varphi(x) := x^2 + cx + c^2$ satisfies the conditions of the theorem. Therefore, we conclude that f is differentiable at $c \in \mathbb{R}$ and that $f'(c) = \varphi(c) = 3c^2$.

We will now establish the Chain Rule. If f is differentiable at c and g is differentiable at $f(c)$, then the Chain Rule states that the derivative of the composite function $g \circ f$ at c is the product $(g \circ f)'(c) = g'(f(c)) \cdot f'(c)$. Note that this can be written as

$$(g \circ f)' = (g' \circ f) \cdot f'.$$

One approach to the Chain Rule is the observation that the difference quotient can be written, when $f(x) \neq f(c)$, as the product

$$\frac{g(f(x)) - g(f(c))}{x - c} = \frac{g(f(x)) - g(f(c))}{f(x) - f(c)} \cdot \frac{f(x) - f(c)}{x - c}.$$

This suggests the correct limiting value. Unfortunately, the first factor in the product on the right is undefined if the denominator $f(x) - f(c)$ equals 0 for values of x near c, and this presents a problem. However, the use of Carathéodory's Theorem neatly avoids this difficulty.

6.1.6 Chain Rule *Let I, J be intervals in $\mathbb{R}$, let $g : I \to \mathbb{R}$ and $f : J \to \mathbb{R}$ be functions such that $f(J) \subseteq I$, and let $c \in J$. If f is differentiable at c and if g is differentiable at $f(c)$, then the composite function $g \circ f$ is differentiable at c and*

$$(g \circ f)'(c) = g'(f(c)) \cdot f'(c). \tag{11}$$

Proof. Since $f'(c)$ exists, Carathéodory's Theorem 6.1.5 implies that there exists a function φ on J such that φ is continuous at c and $f(x) - f(c) = \varphi(x)(x - c)$ for $x \in J$, and where $\varphi(c) = f'(c)$. Also, since $g'(f(c))$ exists, there is a function ψ defined on I such that ψ is continuous at $d := f(c)$ and $g(y) - g(d) = \psi(y)(y - d)$ for $y \in I$, where $\psi(d) = g'(d)$. Substitution of $y = f(x)$ and $d = f(c)$ then produces

$$g(f(x)) - g(f(c)) = \psi(f(x))(f(x) - f(c)) = [(\psi \circ f(x)) \cdot \varphi(x)](x - c)$$

for all $x \in J$ such that $f(x) \in I$. Since the function $(\psi \circ f) \cdot \varphi$ is continuous at c and its value at c is $g'(f(c)) \cdot f'(c)$, Carathéodory's Theorem gives (11). Q.E.D.

If g is differentiable on I, if f is differentiable on J, and if $f(J) \subseteq I$, then it follows from the Chain Rule that $(g \circ f)' = (g' \circ f) \cdot f'$, which can also be written in the form $D(g \circ f) = (Dg \circ f) \cdot Df$.

6.1.7 Examples **(a)** If $f : I \to \mathbb{R}$ is differentiable on I and $g(y) := y^n$ for $y \in \mathbb{R}$ and $n \in \mathbb{N}$, then since $g'(y) = ny^{n-1}$, it follows from the Chain Rule 6.1.6 that

$$(g \circ f)'(x) = g'(f(x)) \cdot f'(x) \quad \text{for} \quad x \in I.$$

Therefore we have $(f^n)'(x) = n(f(x))^{n-1}f'(x)$ for all $x \in I$ as was seen in (9).

(b) Suppose that $f : I \to \mathbb{R}$ is differentiable on I and that $f(x) \neq 0$ and $f'(x) \neq 0$ for $x \in I$. If $h(y) := 1/y$ for $y \neq 0$, then it is an exercise to show that $h'(y) = -1/y^2$ for $y \in \mathbb{R}$, $y \neq 0$. Therefore we have

$$\left(\frac{1}{f}\right)'(x) = (h \circ f)'(x) = h'(f(x))f'(x) = -\frac{f'(x)}{(f(x))^2} \quad \text{for} \quad x \in I.$$

(c) The absolute value function $g(x) := |x|$ is differentiable at all $x \neq 0$ and has derivative $g'(x) = \operatorname{sgn}(x)$ for $x \neq 0$. (The signum function is defined in Example 4.1.10(b).) Though sgn is defined everywhere, it is not equal to g' at $x = 0$ since $g'(0)$ does not exist.

Now if f is a differentiable function, then the Chain Rule implies that the function $g \circ f = |f|$ is also differentiable at all points x where $f(x) \neq 0$, and its derivative is given by

$$|f|'(x) = \operatorname{sgn}(f(x)) \cdot f'(x) = \begin{cases} f'(x) & \text{if} \quad f(x) > 0, \\ -f'(x) & \text{if} \quad f(x) < 0. \end{cases}$$

If f is differentiable at a point c with $f(c) = 0$, then it is an exercise to show that $|f|$ is differentiable at c if and only if $f'(c) = 0$. (See Exercise 7.)

For example, if $f(x) := x^2 - 1$ for $x \in \mathbb{R}$, then the derivative of its absolute value $|f|(x) = |x^2 - 1|$ is equal to $|f|'(x) = \operatorname{sgn}(x^2 - 1) \cdot (2x)$ for $x \neq 1, -1$. See Figure 6.1.1 for a graph of $|f|$.

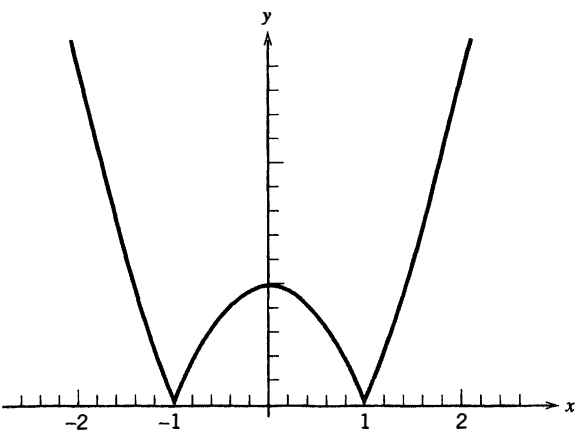


Figure 6.1.1 The function $|f|(x) = |x^2 - 1|$.

(d) It will be proved later that if $S(x) := \sin x$ and $C(x) := \cos x$ for all $x \in \mathbb{R}$, then

$$S'(x) = \cos x = C(x) \quad \text{and} \quad C'(x) = -\sin x = -S(x)$$

for all $x \in \mathbb{R}$. If we use these facts together with the definitions

$$\tan x := \frac{\sin x}{\cos x}, \quad \sec x := \frac{1}{\cos x},$$

for $x \neq (2k+1)\pi/2$, $k \in \mathbb{Z}$, and apply the Quotient Rule 6.1.3(d), we obtain

$$D\tan x = \frac{(\cos x)(\cos x) - (\sin x)(-\sin x)}{(\cos x)^2} = (\sec x)^2,$$

$$D\sec x = \frac{0 - 1(-\sin x)}{(\cos x)^2} = \frac{\sin x}{(\cos x)^2} = (\sec x)(\tan x)$$

for $x \neq (2k+1)\pi/2, k \in \mathbb{Z}$.

Similarly, since

$$\cot x := \frac{\cos x}{\sin x}, \qquad \csc x := \frac{1}{\sin x}$$

for $x \neq k\pi$, $k \in \mathbb{Z}$, then we obtain

$$D\cot x = -(\csc x)^2 \qquad \text{and} \qquad D\csc x = -(\csc x)(\cot x)$$

for $x \neq k\pi$, $k \in \mathbb{Z}$.

(e) Suppose that f is defined by

$$f(x) := \begin{cases} x^2 \sin(1/x) & \text{for } x \neq 0, \\ 0 & \text{for } x = 0. \end{cases}$$

If we use the fact that $D\sin x = \cos x$ for all $x \in \mathbb{R}$ and apply the Product Rule 6.1.3(c) and the Chain Rule 6.1.6, we obtain (why?)

$$f'(x) = 2x\sin(1/x) - \cos(1/x) \quad \text{for} \quad x \neq 0.$$

If $x = 0$, none of the calculational rules may be applied. (Why?) Consequently, the derivative of f at $x = 0$ must be found by applying the definition of derivative. We find that

$$f'(0) = \lim_{x\to 0} \frac{f(x) - f(0)}{x - 0} = \lim_{x\to 0} \frac{x^2 \sin(1/x)}{x} = \lim_{x\to 0} x\sin(1/x) = 0.$$

Hence, the derivative f' of f exists at all $x \in \mathbb{R}$. However, the function f' does not have a limit at $x = 0$ (why?), and consequently f' is discontinuous at $x = 0$. Thus, a function f that is differentiable at every point of $\mathbb{R}$ need not have a continuous derivative f'. □

Inverse Functions

We will now relate the derivative of a function to the derivative of its inverse function, when this inverse function exists. We will limit our attention to a continuous strictly monotone function and use the Continuous Inverse Theorem 5.6.5 to ensure the existence of a continuous inverse function.

If f is a continuous strictly monotone function on an interval I, then its inverse function $g = f^{-1}$ is defined on the interval $J := f(I)$ and satisfies the relation

$$g\big(f(x)\big) = x \quad \text{for} \quad x \in I.$$

If $c \in I$ and $d := f(c)$, and if we knew that both $f'(c)$ and $g'(d)$ exist, then we could differentiate both sides of the equation and apply the Chain Rule to the left side to get $g'\big(f(c)\big) \cdot f'(c) = 1$. Thus, if $f'(c) \neq 0$, we would obtain

$$g'(d) = \frac{1}{f'(c)}.$$